# NUMERICAL METHODS FOR PDES
Lecture 9

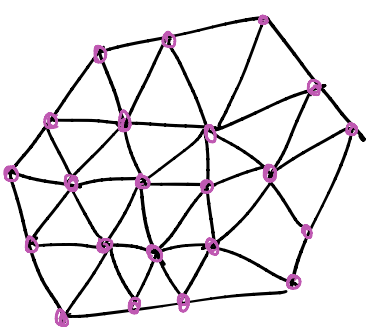
## The FEM in any dim.

Ingredients:

1. $\mathcal{T}_h = \{T\}$ $\forall h$
   (simplices)

triangulation of $\Omega$, so $\bar{\Omega} = \bigcup T$

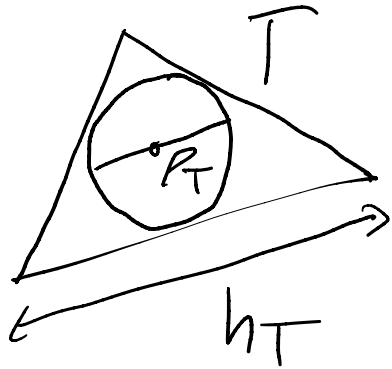
regular (conforming)



$h_T = \text{diam}(T)$

$h = \max_{T \in \mathcal{T}_h} h_T$

$\rho_T$ = diameter of largest ball inside $T$




The family $\mathcal{T}_h$ is SHAPE REGULAR, if

$$\exists \sigma > 0 \ : \ \frac{h_T}{\rho_T} \leq \sigma \quad \forall T \in \mathcal{T}_h$$

2. $$V_h^k = \{ v \in C(\Omega) : v_{|T} \in \mathbb{P}^k(T), \forall T \in \mathcal{T}_h \}$$

$k \in \mathbb{N}$

$$V_{h,0}^k = \{ v \in V_h^k : v_{|\partial\Omega} = 0 \}$$

Prop: $v \in H^1(\Omega) \Leftrightarrow$ ① $v_{|T} \in H^1(T)$ $\forall T$
② if $F = T_1 \cap T_2$, $\exists T_1, T_2$ then $(v_{|T_1})_{|F} = (v_{|T_2})_{|F}$

(3.) Construction of $V_h^k$ by fixing appropriately a set of Lagrangian NODES, e.g.

$\forall T$,

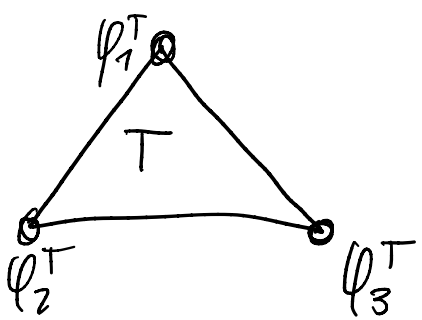
local nodes (if $F = T_1 \cap T_2$ nodes on F will be identified)

$$\mathcal{N}_k(T) = \left\{ x \in T : \; x = \frac{1}{k} \sum_{i=0}^{d} \alpha_i a_i, \; \alpha_i \in \mathbb{N}_0, \; \sum_{i=0}^{d} \alpha_i = k \right\}$$

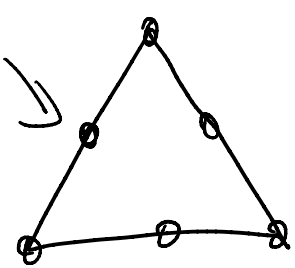
where $\{a_i\}_{i=0}^{d}$ are the vertices of $T$.

Examples:

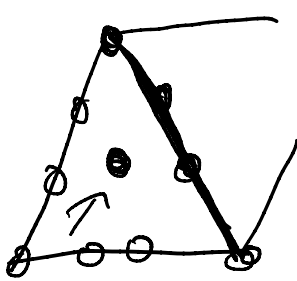
$k=1$




$k=2$



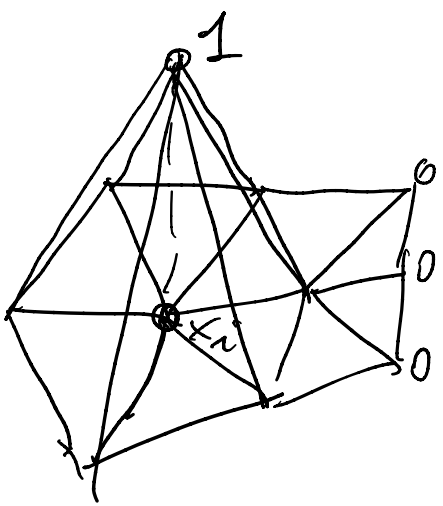
$k=3$



$$\dim \mathbb{P}^k = \binom{d+k}{k}$$

$\Rightarrow$ Lagrangian basis: $\{\varphi_i\}_{i=1}^{\dim \mathbb{P}^k}$

s.t. $\varphi_i(x_j) = \delta_{ij} \quad \forall x_j \in N_k(T)$




$\text{supp}(\varphi_i)$ is located around the supporting node

$\Rightarrow$ FEM system is SPARSE

# FEM ASSEMBLY

FEM setup depends on computing

$$A_{ij} = \int_\Omega k \nabla\varphi_j \cdot \nabla\varphi_i$$

"stiffness" matrix entry

$$\approx \sum_T \int_T k \nabla\varphi_U \cdot \nabla\varphi_i$$

i,j global DOF (node) indices

It is more efficient to assemble by unique loop over TETs ① compute

$$\left(A_{rs} = \left(\int_T k \nabla\varphi_r \cdot \nabla\varphi_s\right)_{rs}\right)$$

r,s local DoF indices

(2) distribute contributions to global stiffness matrix

if $g_T(r)$ map for local-to-global numbering, then

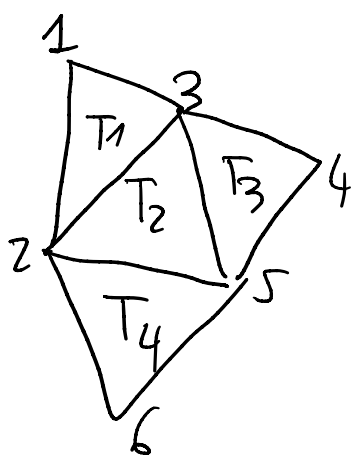
$$A_{g_T(r),g_T(s)} \mathrel{+}= A_{rs}$$

example: linear FEM ($k=1$) $d=2$

connectivity

$$T = \begin{bmatrix} 1 & 2 & 3 \\ \boxed{2 & 5 & 3} \\ 3 & 5 & 4 \\ 2 & 6 & 5 \end{bmatrix}$$

← global numbering of $T_2$ nodes

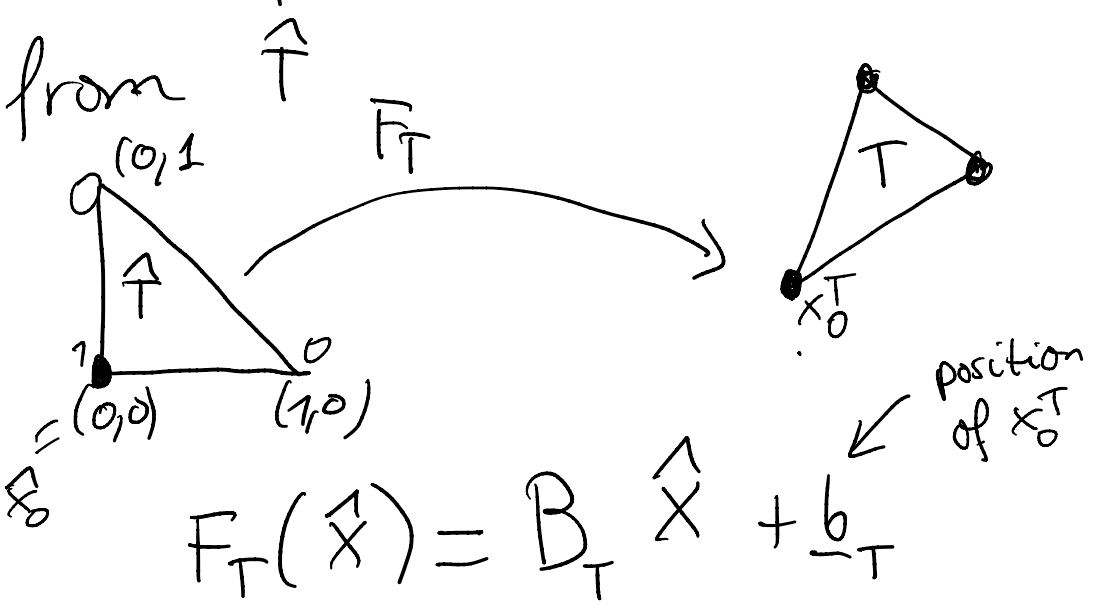


$$\Pi = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \\ \vdots & \\ x_6 & y_6 \end{bmatrix}$$

How to compute $\int_T$ efficiently?

## Reference element

$\forall T \in \mathcal{T}_h$ is image of affine map from $\hat{T}$




$$F_T(\hat{x}) = B_T \hat{x} + \underline{b}_T$$

Jacobian of $F_T$

$F_T$ is invertible and maps the corresponding Lagrange basis of $\hat{T}$ in that of $T$.

$$\varphi_j(x) = F_T \circ \hat{\varphi}_j(\hat{x})$$

$$\int_T K(x) \nabla \varphi_j(x) \cdot \nabla \varphi_i(x) dx$$

$$= \int_{\hat{T}} K\left(F_T(\hat{x})\right) B_T^{-T} \hat{\nabla} \hat{\varphi}_j \cdot B_T^{-T} \hat{\nabla} \hat{\varphi}_i \underbrace{|\det B_T| \, d\hat{x}}_{d\hat{x}}$$

$$\nabla \varphi(x) = B_T^{-T} \hat{\nabla} \hat{\varphi}(\hat{x})$$

$$\int_{\hat{T}} g(\hat{x})\, d\hat{x}$$

$$\approx \sum_{\ell} \omega_\ell\, g(q_\ell)$$

some quadrature formula of the reference element.

## FEM A PRIORI ANALYSTS

Assume

$$= |u - u_h|_1 \le C \inf_{v_h \in V_h^k} \|u - v_h\|_1$$

$$= \|\nabla(u - u_h)\|_0 \qquad \text{(Céa lemma)}$$

$$\leq C \| u - I_h^k u \|_1$$

Interpolant

Def (Lagrange interpolant):

$$I_h^k : C(\bar{\Omega}) \longrightarrow V_h^k$$

$$v \longrightarrow I_h^k v := \sum_j v(x_j)\varphi_j(x)$$

$j$: nodes

where $\{\varphi_j\}$ Lagrange basis w.r.t the FEM nodes $\{x_j\}$

Locally $I_h^k v_{|T} \in \mathbb{P}^k(T)$

Lemma 1 (scaling): For $m \geq 0$ and $v \in H^m(T)$, let $\hat{v} = v \circ F_T$, then

- $\hat{v} \in H^m(\hat{T})$
- $\exists c = c(m) > 0$ :

induced Euclidean norm in $\mathbb{R}^d$

- $|\hat{v}|_{H^m(\hat{T})} \leq c \|B_T\|^m |\det B_T|^{-1/2} |v|_{H^m(T)}$
- $|v|_{H^m(T)} \leq c \|B_T^{-1}\|^m |\det B_T|^{1/2} |\hat{v}|_{H^m(\hat{T})}$

Proof: Q, Lemma 4.2

Lemma 2: we have

- $\|B_T\| \leq h_T/\hat{\rho} \sim h_T$
- $\|B_T^{-1}\| \leq \hat{h}/\rho_T \sim \rho_T^{-1}$

Lemma 3 (Bramble-Hilbert): Let

$$\hat{L} : H^{r+1}(\hat{T}) \longrightarrow H^{m}(\hat{T})$$

linear and continuous for $r, m \geq 0$ such that

$$\hat{L}(\hat{p}) = 0 \qquad \forall \hat{p} \in \mathbb{P}_r(\hat{T})$$

Then for each $\hat{v} \in H^{r+1}(\hat{T})$, we have

$$|\hat{L}(\hat{v})|_{H^m(\hat{T})} \leq C \inf_{\hat{p} \in \mathbb{P}_r(\hat{T})} \|\hat{v} + \hat{p}\|_{H^{r+1}(\hat{T})}$$

$C$ = norm of $\hat{L}$

example: $\hat{L}(\hat{v}) = \hat{v} - \hat{I}\hat{v}$

↳ reference element interpolant

Lemma 4 (Deny–Lions): For each $r \geq 0$

$\exists C = C(r, \hat{T})$ :

$$\inf_{\hat{p} \in \mathbb{P}_r(\hat{T})} \|\hat{v} + \hat{p}\|_{H^{k+1}(\hat{T})} \leq C |\hat{v}|_{H^{r+1}(\hat{T})}$$

$\forall \hat{v} \in H^{r+1}(\hat{T})$

Corollary (Lemma 3 + Lemma 4): "Bramble–Hilbert"

$$|\hat{L}(\hat{v})|_{H^m(\hat{T})} \leq C \, |\hat{v}|_{H^{r+1}(\hat{T})}$$

**Theorem:** (local interpolation)

Let $\boxed{r \geq 1}$, $0 \leq m \leq r+1$. Then $\exists$

$C = C(r, m, \hat{T}) > 0$ :

$$|v - I_h^r v|_{H^m(T)} \leq C \frac{h_T^{r+1}}{\rho_T^m} |v|_{H^{r+1}(T)}$$

$\forall v \in H^{r+1}(T)$

**Proof:**

1) $r \geq 1 \Rightarrow v \in H^2(T) \Rightarrow v \in C(T)$

$\Rightarrow I_h^r v|_T$ makes sense.

$$|v - I_h^r v|_{H^m(T)} \overset{\text{lemma 1}}{\leq} C \|B_T^{-1}\|^m |\det B_T|^{1/2} |\hat{v} - \hat{I}_r \hat{v}|_{H^m(\hat{T})}$$

$$\overset{\text{lemma 2}}{\leq} C \rho_T^{-m} |\det B_T|^{1/2} |\hat{v} - \hat{I}_r \hat{v}|_{H^m(\hat{T})}$$

Corollary of Lemma 3+4

$$\leq C \rho_T^{-m} |\det B_T|^{1/2} |\hat{v}|_{H^{r+1}(\hat{T})}$$

Lemma 1

$$\leq C \rho_T^{-m} |\det B_T|^{1/2} \|B_T\|^{r+1} |\det B_T|^{-1/2} |v|_{H^{r+1}(T)}$$

Lemma 2

$$\leq C \rho_T^{-m} h_T^{r+1} |v|_{H^{r+1}(T)}$$

□

Under ass. of shape regularity

$$\frac{h_T^{r+1}}{\rho_T^m} \leq \underset{\| }{C} \; h^{r+1-m}$$

$O^m$

Corollary: If $\mathcal{T}_h$ is shape-regular

family, then $\forall T \in \mathcal{T}_h$

$$|v - I_h^r v|_{H^m(T)} \leq C h^{r+1-m} |v|_{H^{r+1}(T)}$$

$$C = C(r, m, \hat{T}, \sigma)$$

Theorem (Global interpolation)

Let $m = 0, 1$ and $r \geq 1$. There exists $C = C(r, m, \hat{K}, \sigma)$

$$|v - I_h^r v|_{H^m(\Omega)} \leq C \left( \sum_{T \in \mathcal{T}_h} h_T^{2(r+1-m)} |v|^2_{H^{r+1}(T)} \right)^{1/2}$$

$\forall v \in H^{r+1}(\mathcal{T}_h) \cap H^2(\Omega)$

$$\left[ \leq C h^{r+1-m} |v|_{H^{r+1}(\Omega)}, \ \forall v \in H^{r+1}(\Omega) \right]$$

Using this into Céa gives for $u_h \in V_h^k$ FEM approx

$$|u - u_h|_1 \leq C h^k |u|_{H^{k+1}(T)}$$